Management Performance:

Ene 19 Energy Consumption

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Contributes to 50 credits available for operational energy rating

No Minimum Standard

Aim

To minimise operational energy consumption and the associated carbon emissions by employing effective energy management practices, increasing intrinsic energy efficiency of the building fabric and installing energy efficient building services and equipment.

Question

Annually, how much energy is consumed by the asset in kWh per year (as metered in the reporting period)? Please enter the start and end date of the annual reporting period for consumption data.

Credits	Fuel Type	Units	Start date	End date
-	Electricity	kWh/year		
-	Burning oil	kWh/year		
-	Coal (domestic)	kWh/year		
-	Coking Coal	kWh/year		
-	Diesel	kWh/year		
-	Fuel Oil	kWh/year		
-	Gas Oil	kWh/year		
-	LNG	kWh/year		
-	LPG	kWh/year		
-	Natural Gas	kWh/year		
-	Other Petroleum Gas	kWh/year		
-	Petrol	kWh/year		
-	Petroleum Coke	kWh/year		
-	Refinery Miscellaneous	kWh/year		

Credits	Fuel Type	Units	Start date	End date
-	Biodiesels	kWh/year		
-	Landfill gas	kWh/year		
-	Charcoal	kWh/year		
-	Other biogas	kWh/year		
-	Wood or wood waste	kWh/year		
-	Renewable heat or cooling	kWh/year		
-	District heating	kWh/year		
-	District cooling	kWh/year		

Assessment criteria

Criterion	Assessment criteria
1.	For all energy sources, all consumption data must relate specifically to the floor area of the asset that is being assessed. This must be the GIA floor area that has been entered under Asset Dimensions.
2.	Assessors must ensure that the reporting period for each energy source is between 11 – 13 months (334 to 397 days) and that the maximum time gap between the earliest start and the latest end date across all fuels must be no greater than 14 months (428 days).
3.	All energy consumption must be entered for the asset. If the asset uses an energy source that is not listed, please contact BRE for guidance on how to account for this.

Methodology

Table 33: Fuel specific guidance for calculating energy consumption

Fuel type	Guidance	Notes
Burning oil Diesel Fuel Oil Gas Oil Petrol LNG	Fuel usage can be recorded through: a) Sub-metering of equipment which use these fuels b) Estimates of system efficiency and running times (Oils only) c) Calculations based on inventory data	Please note: the consumption figure is for fuel which has been used directly within the asset NOT for vehicles or other appliances which operate on-site, unless this is specified within the scope.

Fuel type	Guidance	Notes
LPG		
Natural Gas		
Other Petroleum Gas		
Landfill Gas		
Coal (domestic)	Calculating the amount of fuel consumption can be achieved through: a) Sub-metering of equipment which use these fuels b) Invoices for materials purchased during the reporting period and calculations based on the calorific content of the material.	Please note: fuel consumption related to the burning of solid material for the purposes of creating heat, such as Smokeless fuel, Coal and Anthracite.
Coking coal		
Charcoal		
Petroleum Coke		
Biodiesel		Please note: cooking oil can only be used when it has been appropriately refined to a standard which is suitable for fuel usage.
Biogas		Please note: biogas can be used from off-site suppliers or as a result of on-site generation, following a process such as anaerobic digestion etc.
Wood or waste wood	For wood/waste wood, fuel consumption can be achieved through invoices for materials purchased during the reporting period and calculations based on the calorific content of the material.	-
Electricity	Consumption data should be calculated from metered data	-
Renewable Source		
District Heating	For district heating and/or cooling consumption, energy consumption can be calculated through, but not limited to: a) Sub-metering of equipment b) Building energy management systems c) Calculation via energy bills for fuel types if these fuel types are only used for district heating etc.	-
District Cooling		

Note: Where the asset does not have consumption data which covers the assessment area only, the BREEAM In-Use International Energy Allocation Calculator may be used when completing the Management Performance

Energy Category. This can be accessed through the BREEAM In-Use Assessor extranet, or the BREEAM In-Use online platform. Within this tool, there is full guidance explaining the requirements for this method.

Verification of consumption data

Where consumption data has been entered into the Part 2 Energy Calculator, the Assessor must provide evidence to support it and demonstrate its correlation to the assessed area. The consumption figures shall be supported by either corresponding energy bills or verified meter readings. This means that, for example, simply submitting a spreadsheet containing consumption figures is not compliant and clear demonstration that this has been verified is required. Furthermore, it is also possible that the energy bills available for an asset might not coincide with the assessed area for the asset, in which case further evidence is required to demonstrate the assessed consumption for the assessed area.

Since there are many different methods in which energy data can be monitored (e.g. manual meter readings, Energy Management Systems etc.), the format used when submitting meter readings often varies between assessments, leading to confusion for Assessors on whether particular evidence meets the requirements.

When neither energy bills nor meter readings (verified by the Assessor) are available for the assessed area it is possible to submit the following evidence to support consumption data figures:

1. Clear description of how the consumption data was obtained, e.g. manual reading, automatic reading from a BMS, energy bills, etc.;
2. Clear description of any calculation carried out, e.g. energy use determined by subtracting sub-metered data from main meter readings;
3. Clear reporting of the period the consumption data relates to, i.e. start date and end-date for each consumption figure;
4. Clear confirmation that the figures submitted refer to the consumption for the assessed area and that they are based on metered data, rather than estimations or apportionment, except for data submitted using the BREEAM In-Use International Energy Allocation Calculator, see Establishing assessed energy consumption section. Such confirmation will be provided and signed by either someone internal to the organisation or the Assessor themselves.

Note: Where this is undertaken by someone other than the Assessor, the Assessor must be satisfied that the information is correct, and that the individual has a good understanding of metering systems and of the metering strategy of the particular asset being assessed. This should be clearly outlined within the Assessor comments.

Energy sources start being used, or stop being used part way through the reporting period

BREEAM considers the overall consumption for a 12-month period rather than when sources were added or changed during that period. Where an energy source, such as on-site renewable energy generation capacity, begins contributing to or stops contributing to the energy used by the asset part way through the year, the start and end dates entered should still match the reporting period for all energy uses.

Non-standard energy consumption:

The energy consumption for large energy intensive loads and all external lighting may be subtracted from the overall asset energy consumption where it is separately sub-metered.

Examples of Non-standard energy uses include:

a. Regional server room: Energy used in a server room that services multiple satellite offices/servers
b. Trading floor: Energy used in a room where traders are gathered to operate on financial markets

c. Bakery oven: Energy used in a commercial sized oven used to bake foods.

d. Sports floodlighting: Energy used by broad-beamed, high intensity artificial lights (often used outside)

e. Furnace or forming process:

   i. Energy used in a furnace: an industrial devise used for many things, such as extracting metals or as heat source in chemical plants.

   ii. Energy used in forming process: a manufacturing process which makes uses of suitable stresses to cause deformation of materials to produce required shapes.

f. Blast chilling or freezing: Energy used for cooling materials (often food) quickly to low temperatures.

Reporting period

The intention is that the user enters energy consumption data based on a 365-day reporting period; for both years. However, it is possible to enter data for any reporting period between 11 and 13 months. Any reporting period outside of the 11 to 13-month range would be invalid and result in zero credits being scored.

Validity of consumption data

In order to maintain the reliability of data entered into the BREEAM In-Use Online Platform, any consumption-related data submitted must have a reporting period start date which is no more than one year prior to the start of the assessment (creation of a measurement).

For example, if the assessment was started on the 3rd of January 2020, the reporting period start date cannot be prior to 3rd of January 2019. A more recent period can always be entered, as long as the Assessor can verify that the data is correct for the period entered.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
1	Calculations based on inventory data.
2	Copies of energy bills or verified meter readings/ photographic evidence of meters for the beginning and end of the reporting period for each energy source.